MATH329 · OPERATIONS RESEARCH

Formulation Solutions

Six formulation problems from the slides and the Examples file

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01 Solution Q1: Bader

x_1 = number of shirts per week x_2 = number of blouses per week

Available minutes per department = workers × 8 hours × 5 days × 60 minutes.

$$\text{Maximize } z = 8x_1 + 12x_2$$

$$\begin{aligned}
\text{subject to} \quad & 20x_1 + 60x_2 \le 60000 && \text{(cutting: } 25 \times 40 \times 60\text{)} \\
& 70x_1 + 60x_2 \le 84000 && \text{(sewing: } 35 \times 40 \times 60\text{)} \\
& 12x_1 + 4x_2 \le 12000 && \text{(packaging: } 5 \times 40 \times 60\text{)} \\
& x_1, x_2 \ge 0, \text{ integers}
\end{aligned}$$

02 Solution Q2: Wild West

x_1 = number of Type I hats per day x_2 = number of Type II hats per day

Let a Type II hat take c minutes. A Type I hat takes $2c$, and the available time is $400c$. Divide by c:

Maximize $z = 8x_1 + 5x_2$
subject to $2x_1 + x_2 \leq 400$ (labor)
$x_1 \leq 150$ (market limit, Type I)
$x_2 \leq 200$ (market limit, Type II)
$x_1, x_2 \geq 0$, integers

03 Solution Q3: Feed Mix

x_1 = kg of variety 1 x_2 = kg of variety 2

Put the data in a table first, then read each row as a constraint:

Nutrient	Variety 1	Variety 2	At least
A	1	3	15
B	2	2	20
C	3	2	24
Cost (Rs./kg)	2	3	

Min $Z = 2x_1 + 3x_2$
S.T $x_1 + 3x_2 \geq 15$
$2x_1 + 2x_2 \geq 20$
$3x_1 + 2x_2 \geq 24$
$x_1, x_2 \geq 0$

04 Solution Q4: Tonics

x = units of tonic X y = units of tonic Y

Vitamin	Tonic X	Tonic Y	Daily requirement
A	2	4	40
D	3	2	50
Cost (Rs./unit)	5	3	

$$\text{Minimize } Z = 5x + 3y$$

$$\text{S.T} \quad 2x + 4y \geq 40$$

$$3x + 2y \geq 50$$

$$x, y \geq 0$$

05 Solution Q5: Reddy Mikks

x_1 = tons of interior paint per day x_2 = tons of exterior paint per day

The sentence "interior cannot exceed exterior by more than 1" means the difference $x_1 - x_2$ is at most 1:

$$\text{Maximize } z = 5x_1 + 4x_2 \quad \text{(in \$1000)}$$

$$\text{subject to} \quad 6x_1 + 4x_2 \leq 24 \quad \text{(raw material M1)}$$

$$x_1 + 2x_2 \leq 6 \quad \text{(raw material M2)}$$

$$x_1 - x_2 \leq 1 \quad \text{(market limit)}$$

$$x_1 \leq 2 \quad \text{(demand limit)}$$

$$x_1, x_2 \geq 0$$

06 Solution Q6: Trim Loss

The variables here are not the number of tops; they are the number of sheets cut with each pattern:

Pattern	10 cm tops	20 cm tops	Sheets
1	10	0	x_1
2	2	2	x_2
3	6	1	x_3

$$\text{Min } Z = x_1 + x_2 + x_3$$

$$\begin{aligned} \text{subject to} \quad & 10x_1 + 2x_2 + 6x_3 \geq 20000 && \text{(small tops)} \\ & 2x_2 + x_3 \geq 15000 && \text{(large tops)} \\ & x_1, x_2, x_3 \geq 0 \text{ and integers} \end{aligned}$$

The final model is from the slide. I derived the pattern table from its coefficients: a 20 × 50 sheet fits 10 small tops, or 2 large with 2 small, or 1 large with 6 small.